SMS/ROMOLD POLYPROPYLENE (PP) MAINTENANCE HOLE

REV.	DESCRIPTION	DATE	DRAWN
5	SHEET 2 CONFIGS REVISED BACK TO PREVIOUS ANGELS, FEW MORE CHANGES AS PER DISCUSSIONS	02/09/25	SB
4	UPDATED SHEET 5 WITH CLEAR OPENING SIZES	09/05/25	SB
3	NOTES REFINED POST FURTHER REVIEW	21/03/25	SB
2	ADDED NOTE POST LMW REVIEW	14/03/25	SB
1	INTIAL RELEASE	11/03/25	SB

REVISIONS

ROMOLD

Unless Otherwise Stated:
All toleranes as stated
Surface Finish: **0.8µm**
All Dimensions: **mm**

DECIMALS		ANGLES	
X	±1.0	X	±1°
X.X	±0.1	X.X	±0.3°
X.XX	±0.02		

Material:

Finish:

Approx Weight:

Description:
SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE

Part Number:

Drawing Scale: **NTS**

DO NOT SCALE DRAWING IF IN DOUBT ASK

Drawing Number: **MH-ASSY-DN1000-100**

Sheet: **1 of 6**

Revision: **5**

Drawing Produced In Accordance With: **AS 1100**

Projection Method: **THIRD ANGLE**

Sheet Size: **A2**

Drawn by: **SB**

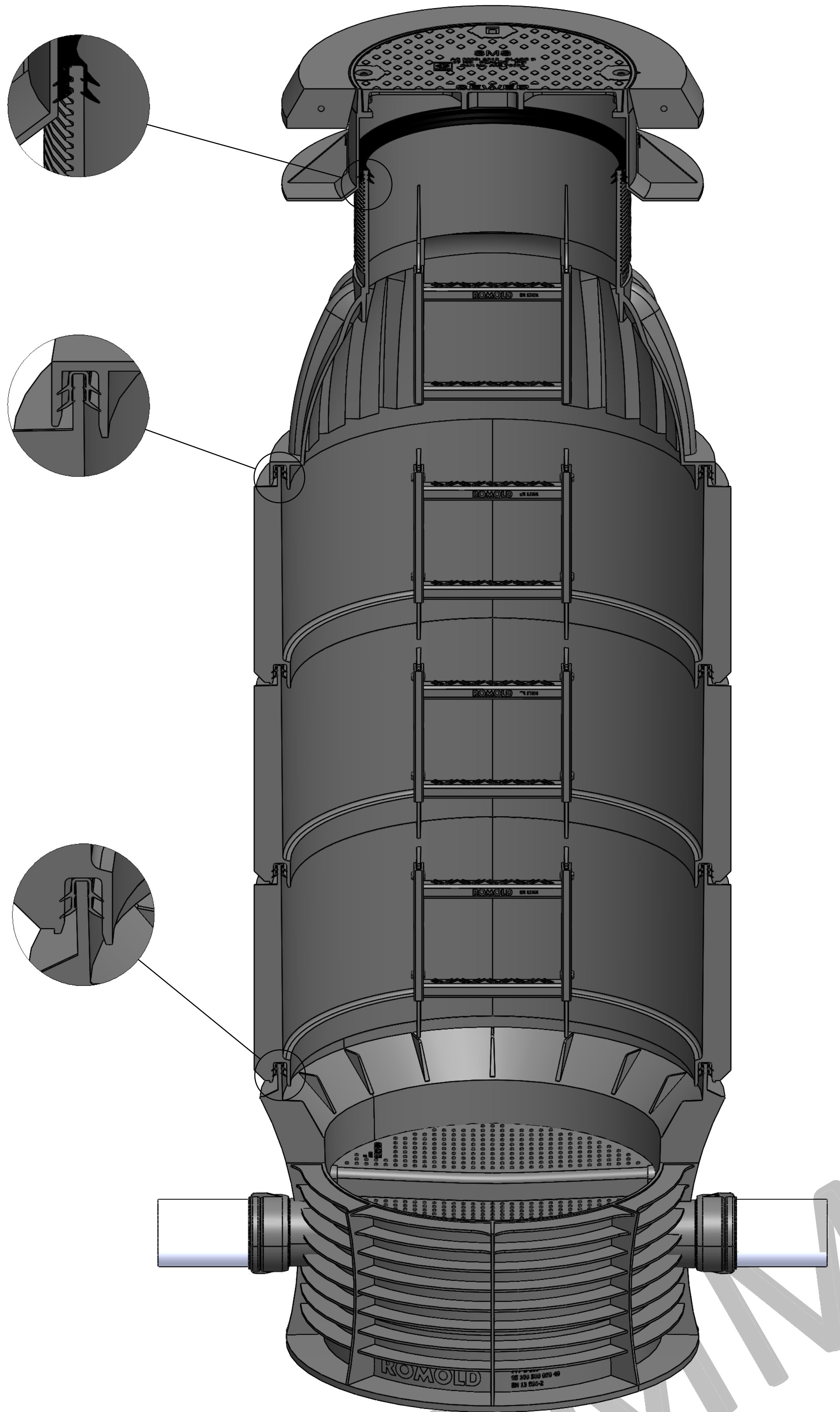
Drawn Date: **11/03/25**

Checked/Approved by: **HB**

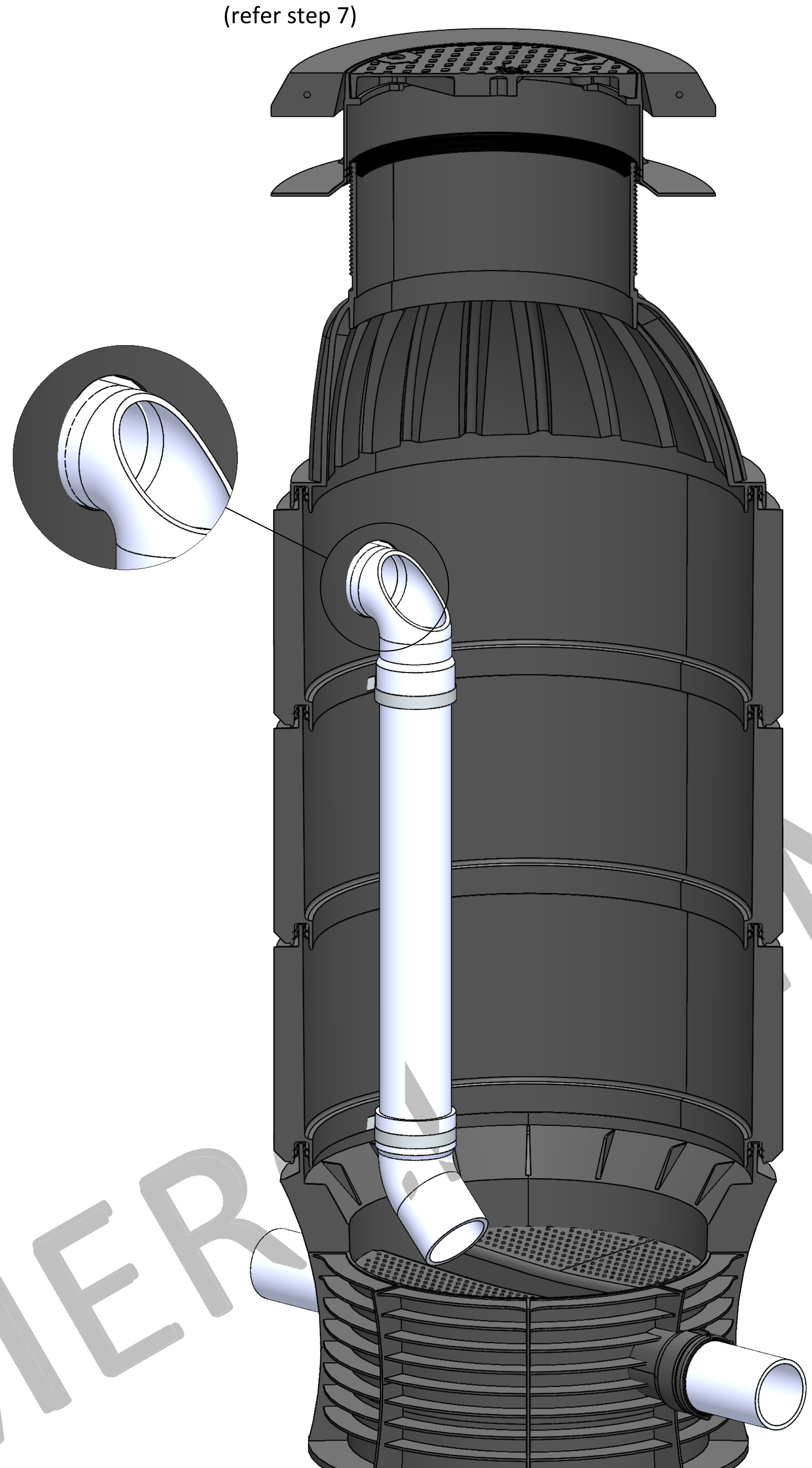
Checked/Approved Date: **11/03/25**

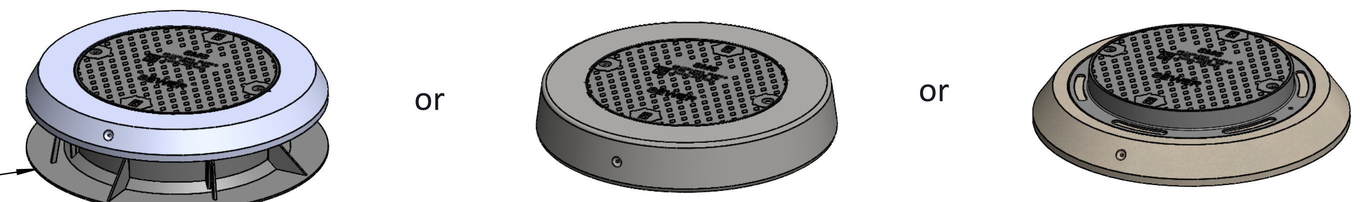
SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE (WITH 600mm CLEAR OPENING ACCESS COVER)
(INCLUDING INSTALLATION GUIDELINES)

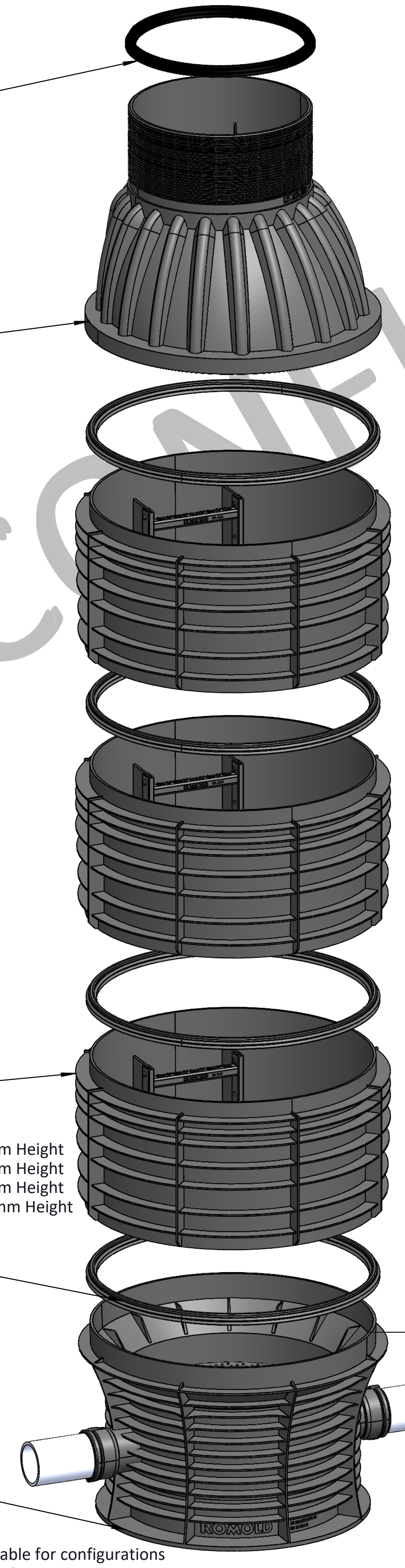
MAINTENANCE HOLE ASSEMBLY



MAINTENANCE HOLE WITH DROP PIPE ASSEMBLY
(refer step 7)



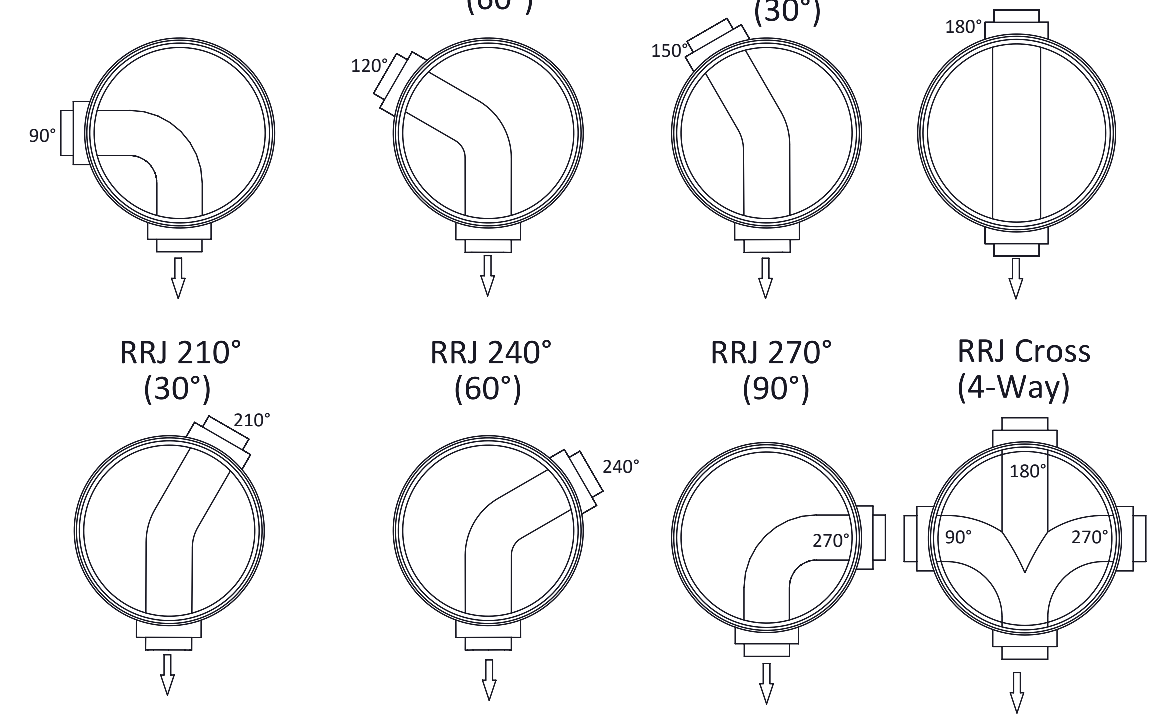




AVAILABLE STANDARD CONFIGURATIONS*

MH Base

CODE	CONFIGURATION	INLET/OUT SIZE
MHB1000150090	RRJ 90°	DN150
MHB1000150120	RRJ 120°	DN150
MHB1000150150	RRJ 150°	DN150
MHB1000150180S	RRJ 180° Straight	DN150
MHB1000150210	RRJ 210°	DN150
MHB1000150240	RRJ 240°	DN150
MHB1000150270	RRJ 270°	DN150
MHB10001504WC	RRJ Cross (4-Way)	DN150
MHB1000225090	RRJ 90°	DN225
MHB1000225120	RRJ 120°	DN225
MHB1000225150	RRJ 150°	DN225
MHB1000225180S	RRJ 180° Straight	DN225
MHB1000225210	RRJ 210°	DN225
MHB1000225240	RRJ 240°	DN225
MHB1000225270	RRJ 270°	DN225
MHB10002254WC	RRJ Cross (4-Way)	DN225
MHB1000300090	RRJ 90°	DN300
MHB1000300120	RRJ 120°	DN300
MHB1000300150	RRJ 150°	DN300
MHB1000300180S	RRJ 180° Straight	DN300
MHB1000300210	RRJ 210°	DN300
MHB1000300240	RRJ 240°	DN300
MHB1000300270	RRJ 270°	DN300
MHB10003004WC	RRJ Cross (4-Way)	DN300
MHB1000375090	RRJ 90°	DN375
MHB1000375120	RRJ 120°	DN375
MHB1000375150	RRJ 150°	DN375
MHB1000375180S	RRJ 180° Straight	DN375
MHB1000375210	RRJ 210°	DN375
MHB1000375240	RRJ 240°	DN375
MHB1000375270	RRJ 270°	DN375
MHB10003754WC	RRJ Cross (4-Way)	DN375



* Other custom configurations available on request

⇩ downstream

INDUSTRY STANDARDS

- WSA 137 – Industry Standard for Unplasticised Poly (vinyl chloride) (PVC-U), Polypropylene (PP), and Polyethylene (PE) Maintenance Shafts, Chambers, and Holes for Sewerage
- EN 13598-2 – Plastics Piping Systems for Non-Pressure Underground Drainage and Sewerage – Specifications for Manholes and Inspection Chambers in Traffic Areas and Deep Underground Installations
- EN 14396 – Fixed Ladders for Manholes
- AS/NZS 2566 for Buried Flexible Pipelines
- AS/NZS 5065 – Polyethylene and Polypropylene Pipes and Fittings for Drainage and Sewerage Applications
- AS 3996 – Access Covers and Grates
- AS 1646 – Elastomeric Seals for Waterworks Purposes

APPLICABLE PRODUCT SPECIFICATIONS

- WSA PS-315 – Fixed Ladders for Man Entry Structures (Water Supply & Sewerage)
- WSA PS-340 – Maintenance Holes (MH) – Polypropylene (PP) for Non-Pressure Sewerage Applications
- WSA PS-290 – Access Covers and Frames for Water Supply and Sewerage (WSA 132 Compliance)
- WSA PS-230 – Unplasticised Polyvinyl Chloride (PVC-U) Pipes and Fittings for Non-Pressure Sewerage and Drainage Applications

SITE SAFETY

Carry out relevant safety assessment/s before carrying out any installation/construction activities.

ROMOLD

Unless Otherwise Stated:	DECIMALS		ANGLES	
All toleranes as stated	X	±1.0	X	±1°
Surface Finish: **0.8µm**	X.X	±0.1	X.X	±0.3°
All Dimensions: **mm**	X.XX	±0.02		

Material:

Finish:

Approx Weight:

Drawing Scale: **NTS**

DO NOT SCALE DRAWING IF IN DOUBT ASK

Drawing Produced In Accordance With: **AS 1100**

Projection Method: **THIRD ANGLE**

Sheet Size: **A2**

sms

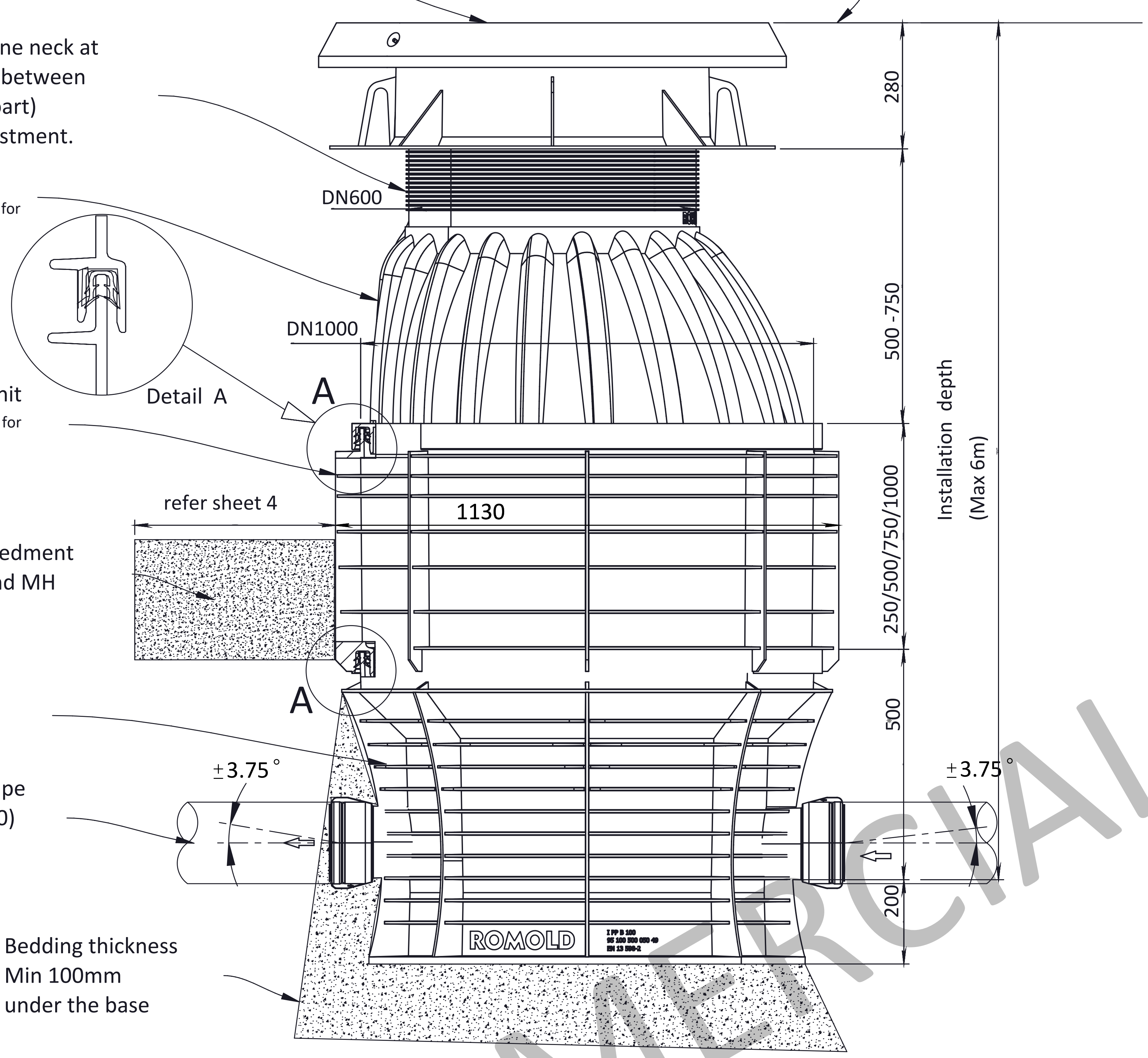
Description: **SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE**

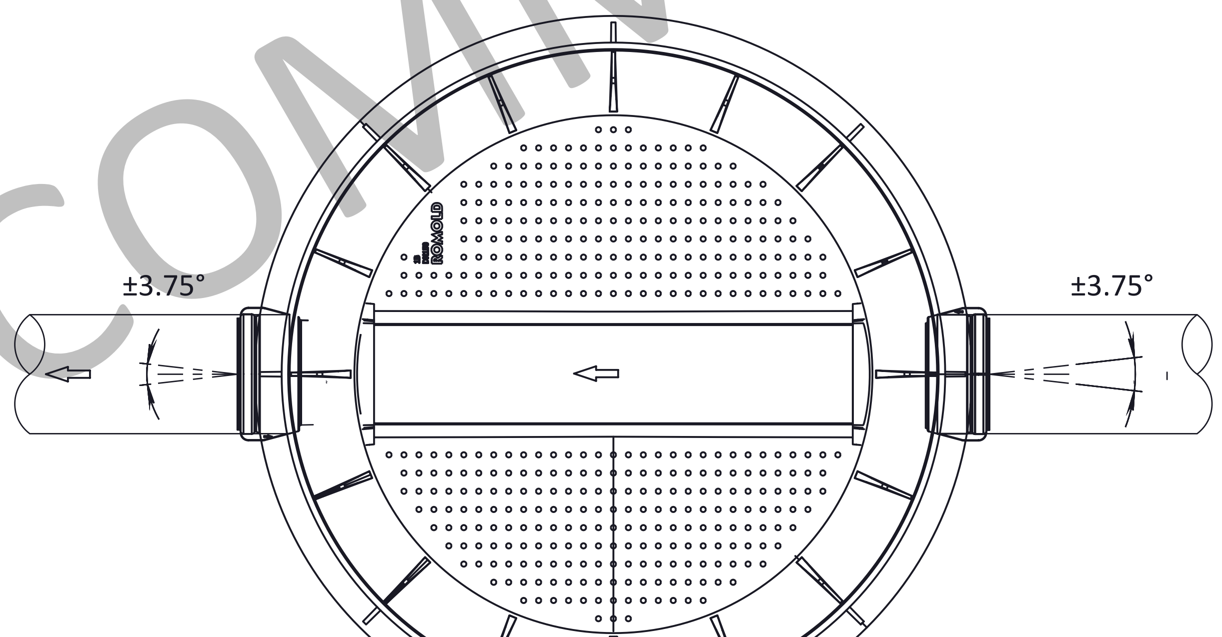
Part Number:

Drawing Number:	Sheet:	Revision:
MH-ASSY-DN1000-100	2 of 6	5

Drawn by:	Drawn Date:
SB	11/03/25
Checked/Approved by:	**Checked/Approved Date:**
HB	11/03/25

ASSEMBLY AND INSTALLATION





GENERAL SAFETY NOTE:
All installation, handling, and maintenance activities must be carried out in strict accordance with relevant local, national, and industry safety standards and regulations. Appropriate personal protective equipment (PPE) must be worn at all times.

1. TRANSPORT AND STORAGE
Store chamber elements vertically on level ground. In case of extended outdoor storage, protection of the chambers against the sun is vital. All supplied element seals have to be stored in their packaging, protected from frost and direct sunlight.

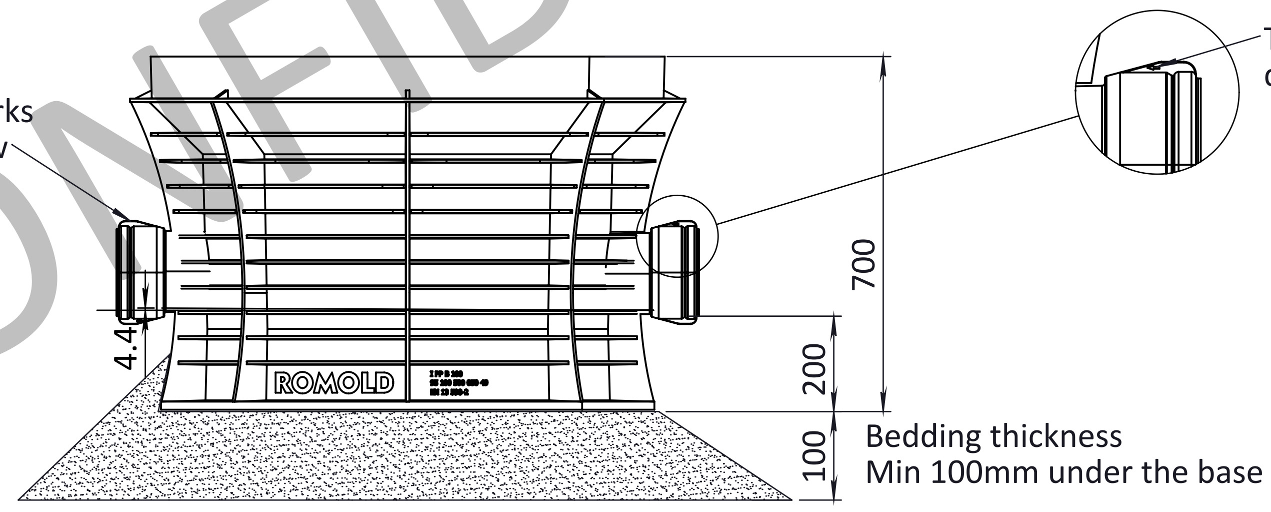
2. GENERAL INFORMATION
SMS/ROMOLD PP maintenance holes are provided ready to connect. Deliveries must be checked for completeness.
All components must be checked for damage or contamination before installation and cleaned or replaced if necessary.
Damaged components must not be installed.

3. ASSEMBLY AND INSTALLATION
All of the installation parameters listed below must be followed, as detailed in this installation guide.

Step 1: BEDDING (GRANULAR SUB-BASE)
The minimum depth required below the base is 100 mm. The support area of the maintenance hole base must be load-bearing and completely levelled.

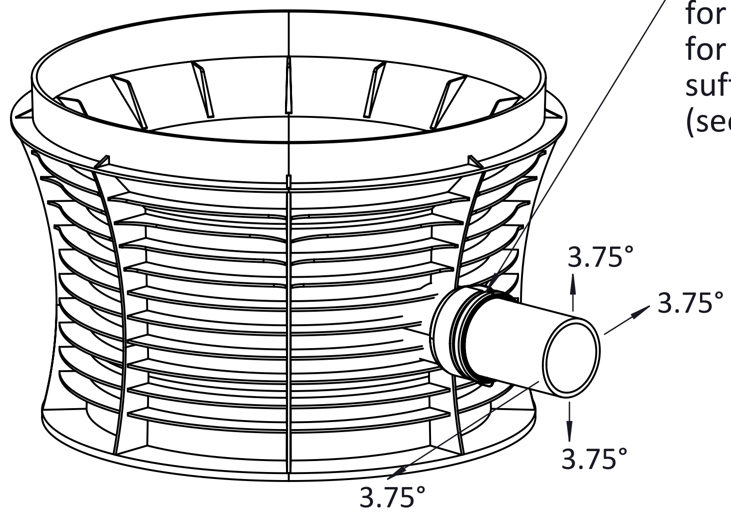
The support area of the maintenance hole base must be established in accordance with the planning (differential between base to channel level = 200 mm).

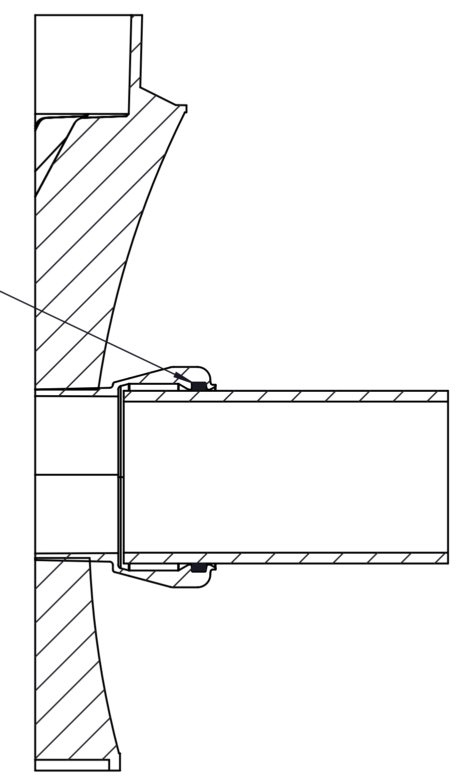
Step 2: BASE/PIPE CONNECTION
The base shall be positioned/levelled on the prepared support area in accordance with the connecting pipes. The adjustment and flow direction of the maintenance hole base must be checked. The adjustment and flow direction of the maintenance hole base must be controlled.



Step 2.1 PP-BASE WITH SOCKETS
The inserted seals should be checked for correct fitment and inspected for any damage, cleaning may be necessary.
Apply sufficient lubricant on the connecting pipe in the socket as well as at the end of the spigot and fully insert the pointed end in the socket. **(Suggested approved lubricant: Thomas Grozier BK Bactericidal Pipe Lubricant.)**
For all base sockets horizontal angles of ± 3.75° and gradient changes up to ± 3.75° are **achievable**. Simultaneous direction and gradient changes will reduce the indicated maximum values accordingly. No connectors (short pipes or joints) are required between SMS/ROMOLD PP maintenance holes and pipes, unless specified by local water agency. If fittings are used, check **minimum** insertion depths and seal position, as per specified by local water agency.

The inserted seal should be checked for correct fittings and inspected for damage, cleaned and apply sufficient lubricant (see section view)

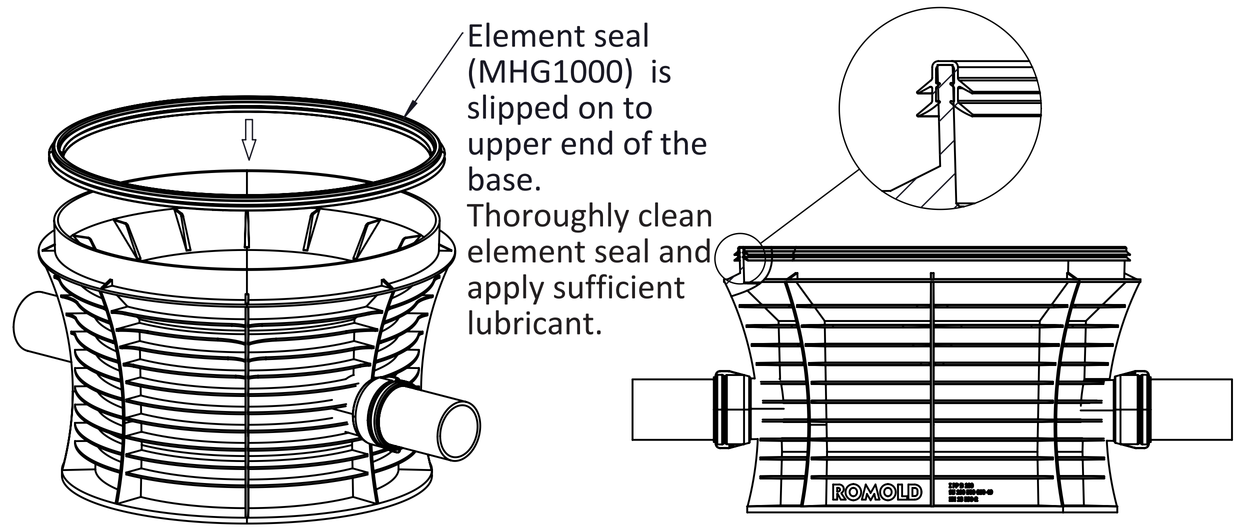




This drawing and any information or descriptive material set out on it are the confidential and copyright property of SMS. © and MUST NOT BE DISCLOSED, COPIED, LOANED in whole or part or used for any purpose without the written permission of **SMS.**	Material:	Description: **SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE**	
	Finish:		
Unless Otherwise Stated: All tolerances as stated Surface Finish: 0.8µm All Dimensions: mm — DECIMALS: X ±1.0, X.X ±0.1, X.XX ±0.02 — ANGLES: X ±1°, X.X ±0.3°	Approx Weight:	Part Number:	
Drawing Scale: NTS	DO NOT SCALE DRAWING IF IN DOUBT ASK	Drawing Number: **MH-ASSY-DN1000-100**	Sheet: **3 of 6** — Revision: **5**
Drawing Produced In Accordance With: **AS 1100**		Drawn by: **SB**	Drawn Date: **11/03/25**
Projection Method: **THIRD ANGLE** — Sheet Size: **A2**	sms	Checked/Approved by: **HB**	Checked/Approved Date: **11/03/25**

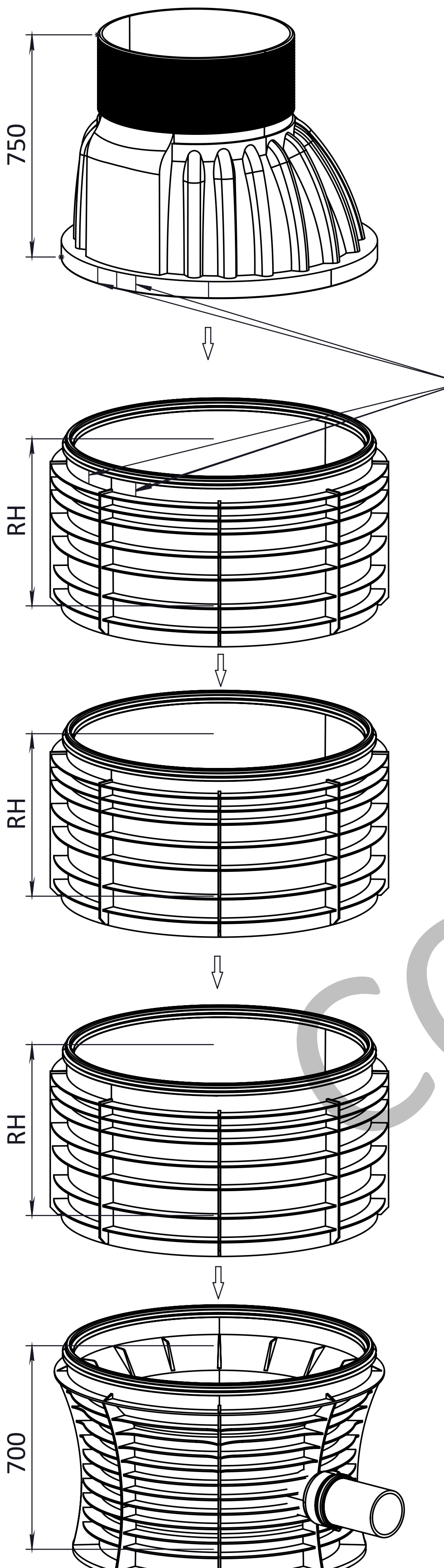
ROMOLD

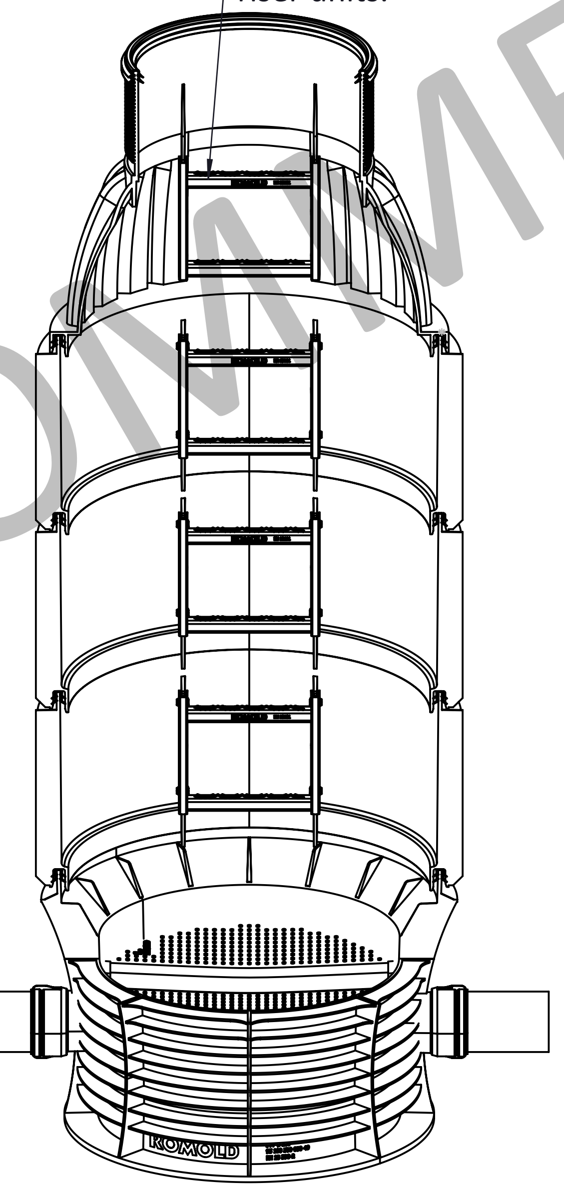
Step. 3 CONNECTION OF MAINTENANCE HOLE ELEMENTS
To achieve each sections connection, the SMS/ROMOLD element seal is to be fitted onto the upper end of the base or ring and checked for precise seating.
Thoroughly clean SMS/ROMOLD seating area and element seal and apply sufficient lubricant.
Clean the slot of the upper element and join/fit together with the element seal to the lower element without tilting. Ensure joint is fully seated.



Calculation of Maintenance Hole height
(Refer Sheet 5 Table 1 &2)

When calculating component requirements (Base, risers, cone and access cover configurations used), ensure that the total height required from the base bottom or pipe invert falls between the min/max adjustment of the access cover configuration used.





Step 4: BACKFILLING MATERIAL
It is important to ensure that non-cohesive, well-graded (all sizes of material), compressible materials are used for backfilling. The maximum particle size of rounded gravel material as a general guide shall not exceed 32 mm and 16 mm, if broken material is used and strictly follow local water agency size requirements.

Step 5: BACKFILLING AND COMPACTING
The width for backfilling around the maintenance hole must be in at any point at least 400 mm. When installing the maintenance holes in hi water table/groundwater conditions (refer local water agency specifications), a backfilling width of at least 500 mm is to be maintained all around to prevent uplift.

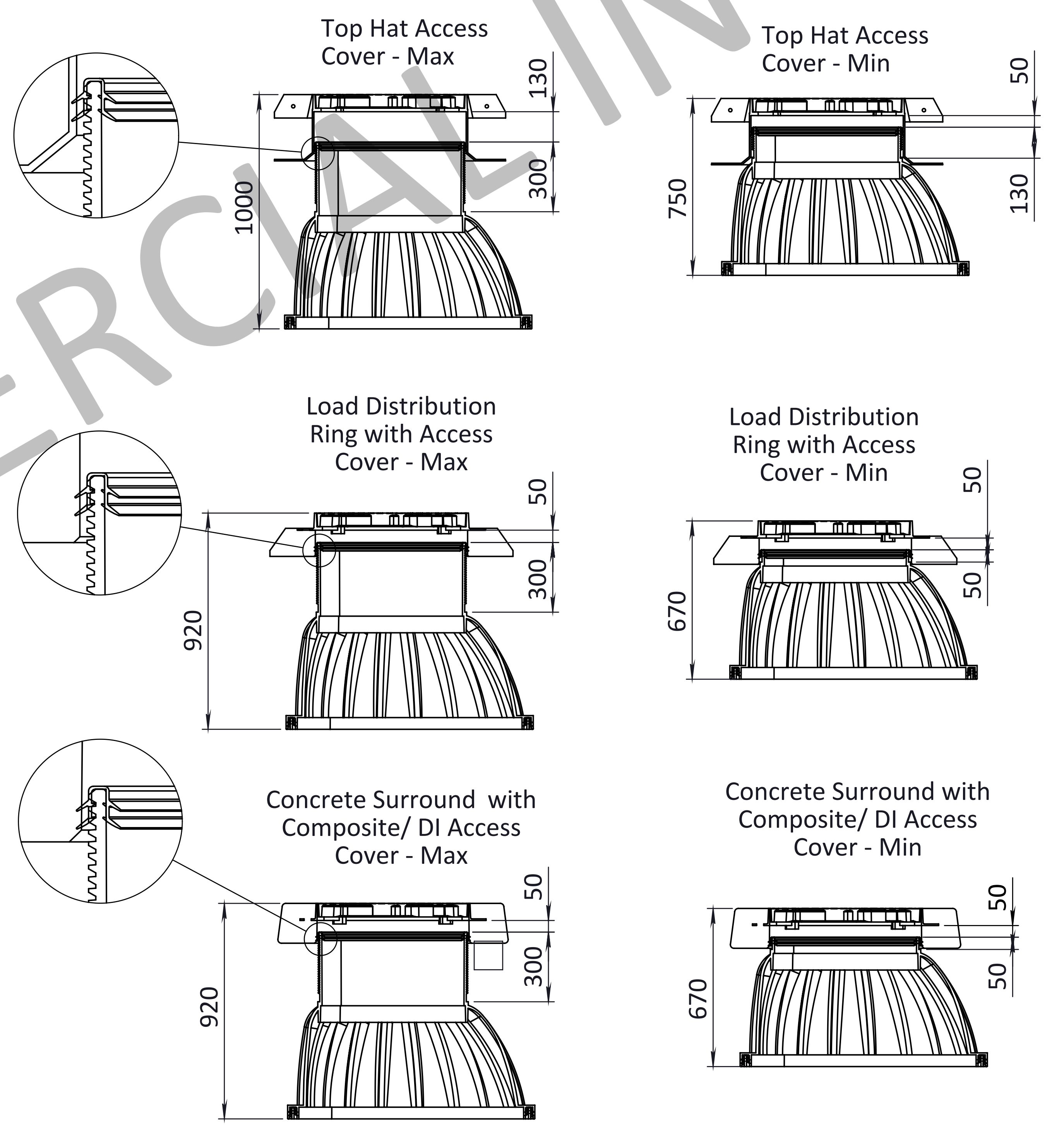
The area of the pipe connection to the maintenance hole has to be carefully under-packed/haunching to ensure correct pipe support e.g. with a narrow hand stamper. The backfilling material is to be inserted carefully and in layers of 200-400 mm layer thickness and compacted with a medium vibrating stamper (approx. 50kg).

The number of required compacting passes per layer depends on the backfilling material. A minimum degree of compaction of DPr = 97 % is to be established for the entire depths of the maintenance hole.

Installation Tip:
Before placing the backfilling material, attach the upper unit (without the seal) to the base or ring. Use a construction-site cover or steel plate to cover the upper unit. Once secured, proceed to pour the backfilling material over the cover, ensuring it is evenly distributed around the maintenance hole. This process protects the maintenance hole from contamination during backfilling.

ALIGNMENT
Align all maintenance hole elements in accordance with vertical marks to ensure the vertical alignment of the ladder. The maintenance hole elements are connected together fully by using body weight or modest force only.

Step 6: HEIGHT ADJUSTMENT
To adjust the height, shorten the neck of the upper unit. SMS/ROMOLD PP maintenance holes can be shortened by a maximum of 250 mm. The cutting is to be done with a saw between two ribs of the cone.



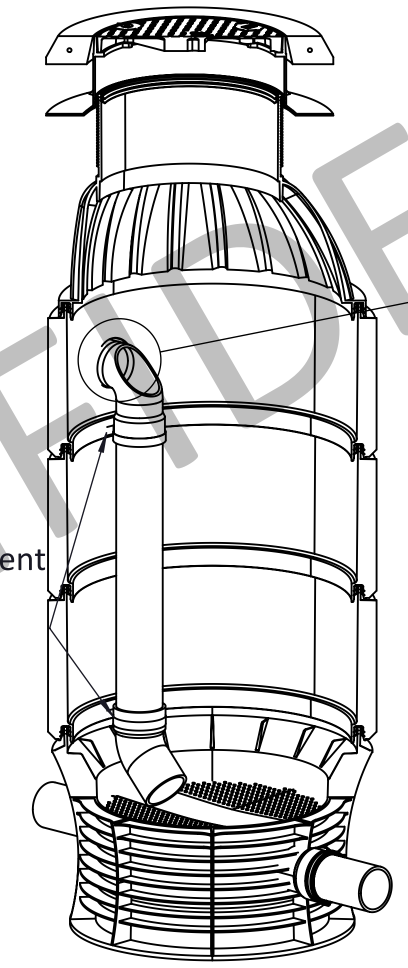
Note: Exercise caution with height adjustment to prevent direct load on the cone from the access cover

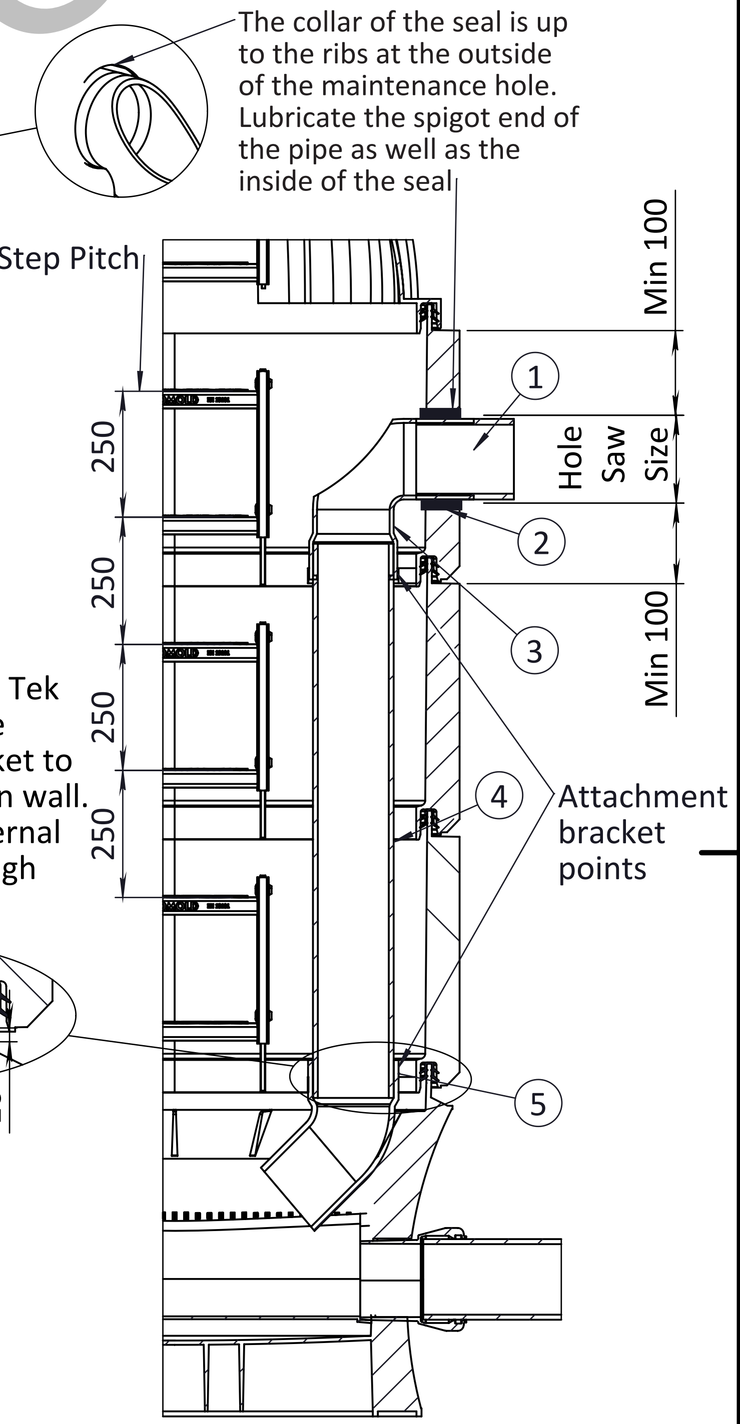
Step 7: DN100/DN150 LATERAL DROP-IN CONNECTIONS (OPTIONAL -INSTALL ON-SITE)
Using an electric hand drill with a hole saw cutter, drill from the inside at the desired location, ensuring that the drilling depth does not exceed the specified maximum limit. Do not drill in any area where a connecting element is present. After drilling, carefully deburr the hole to remove sharp edges and debris. Insert the seal from the outside of the maintenance hole, making sure the collar of the seal sits flush against the ribs on the outer surface.

Lubricate both the spigot end of the pipe and the inner surface of the seal before inserting the pipe. Carefully push the pipe into the seal, ensuring it creates a secure inner overlap for a proper connection. For multiple drop-in connections, repeat same procedure as permitted by local Water Agency, adhering to their specific installation guidelines for such connections.

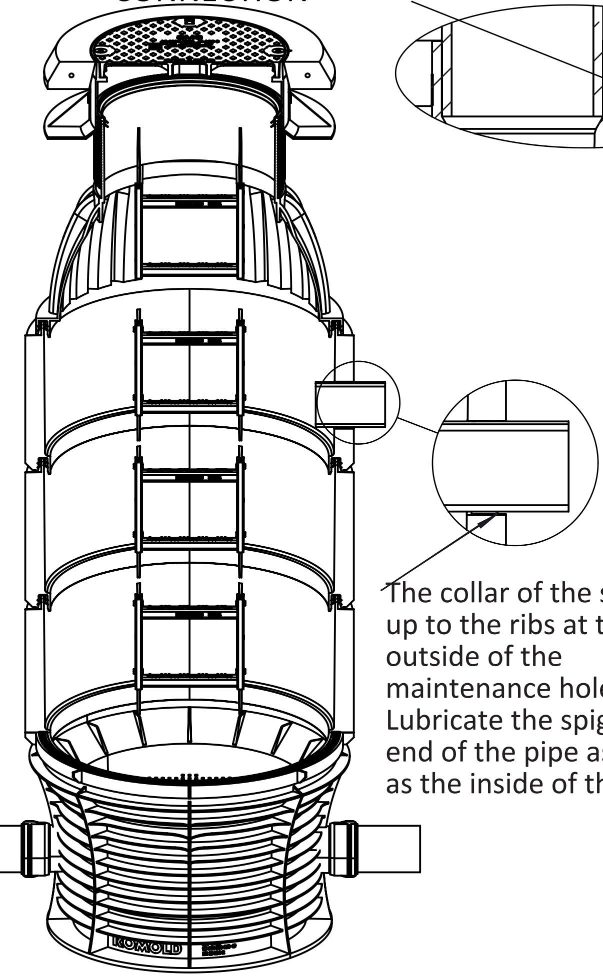
Inlet Pipe (DN)	Lateral Connector SMS Part Number	Hole Saw Cutter SMS Part Number
100	LC5-LAT100MH	LCS100MH
150	LC6-LAT150MH	LCS150MH
200	LC7-LAT200MH	LCS200MH

INTERNAL DROP-IN CONNECTION





EXTERNAL DROP-IN CONNECTION



ITEM NO.	DESCRIPTION	MATERIAL
1	DN100/DN150 Straight Connector	AS/NZS 1260 PVC
2	DN100/DN150 Rubber Collar	AS681.1 Rubber
3	DN100/DN150 90° Elbow	AS/NZS 1260 PVC
4	DN100/DN150 Pipe	AS/NZS 1260 PVC
5	DN100/DN150 45° Elbow	AS/NZS 1260 PVC

ROMOLD	Material:	Finish:	Description: **SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE**
Unless Otherwise Stated: All toleranes as stated; Surface Finish: **0.8μm**; All Dimensions: **mm**	DECIMALS: X ±1.0; X.X ±0.1; X.XX ±0.02	ANGLES: X ±1°; X.X ±0.3°	Approx Weight:
Part Number:			
Drawing Scale: **NTS**	DO NOT SCALE DRAWING IF IN DOUBT ASK	Drawing Number: **MH-ASSY-DN1000-100**	Sheet: **4 of 6** / Revision: **5**
Drawing Produced In Accordance With: **AS 1100**		Drawn by: **SB**	Drawn Date: **11/03/25**
Projection Method: **THIRD ANGLE**	Sheet Size: **A2**	Checked/Approved by: **HB**	Checked/Approved Date: **11/03/25**

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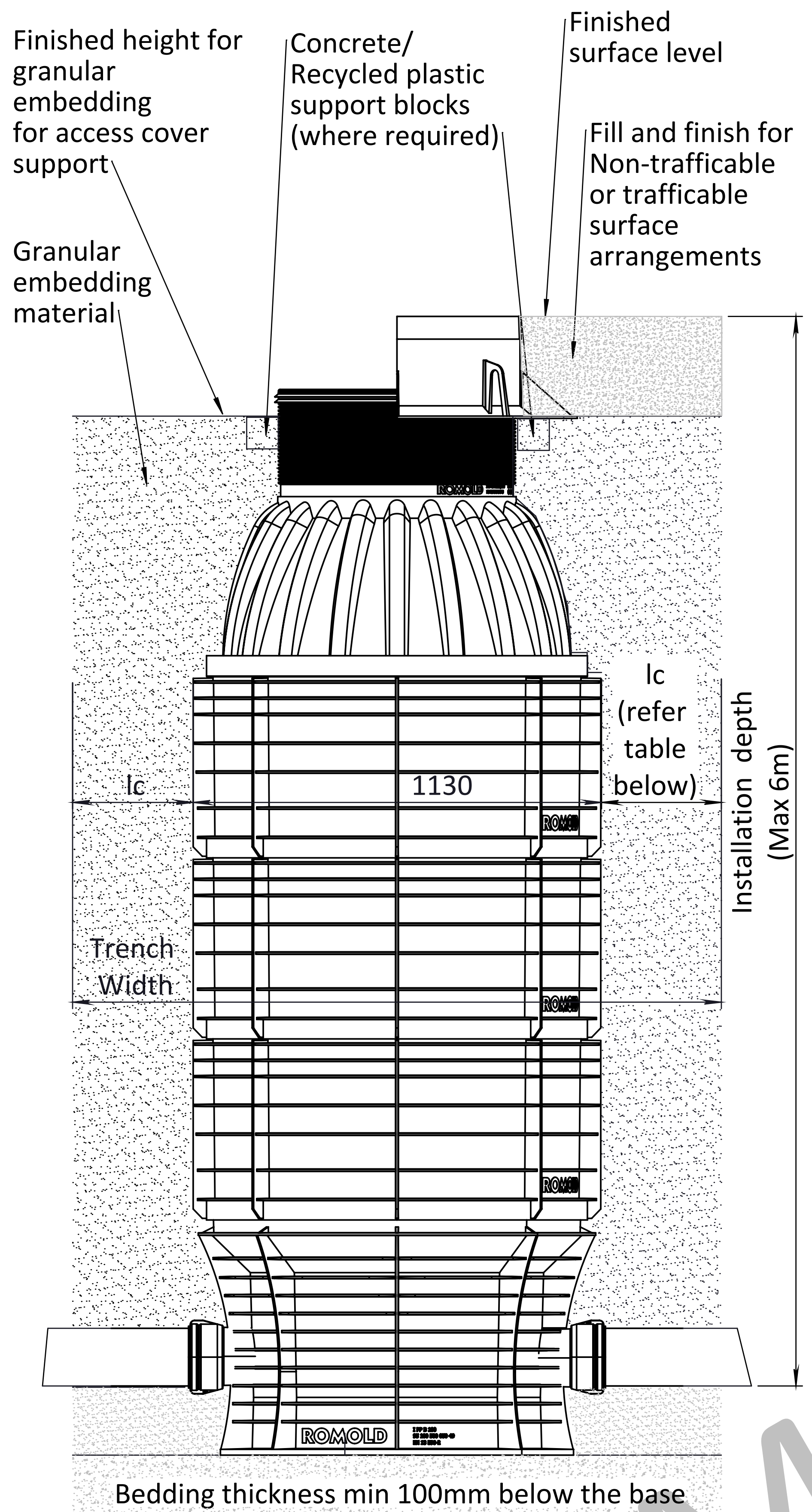
INSTALLATION OVERVIEW
(INCLUDING ACCESS COVER OPTIONS)



Installation notes:

- Refer to local Water Agency specifications for other specific installation guidelines.
- For installation in reactive soils, refer to local water agency installation guidelines.
- Concrete used for installation or concrete components to comply to local water agency specifications.
- Test installed maintenance hole to local water agency requirements.
- Refer to AS/NZS 2566 for Buried Flexible Pipelines and local authority requirements for any specific installation requirements.

Bedding, Backfilling/Embedment material:

- Non-cohesive material rounded gravel up to 32mm broken & gravel up to 16mm width.
- Compaction: DPr ≥ 97%.
- Trench widths

	lc (Min.)	Trench Width (Min.)
Standard Installation	400	1930
Hi Water Table/ Groundwater conditions	500	2130

Installation depth notes:

- Standard Installation max 6m depth, in hi water table/ groundwater conditions max 5m depth of installation.

STANDARD: TOP HAT ACCESS COVER (OPTIONAL VEGETATION RING)

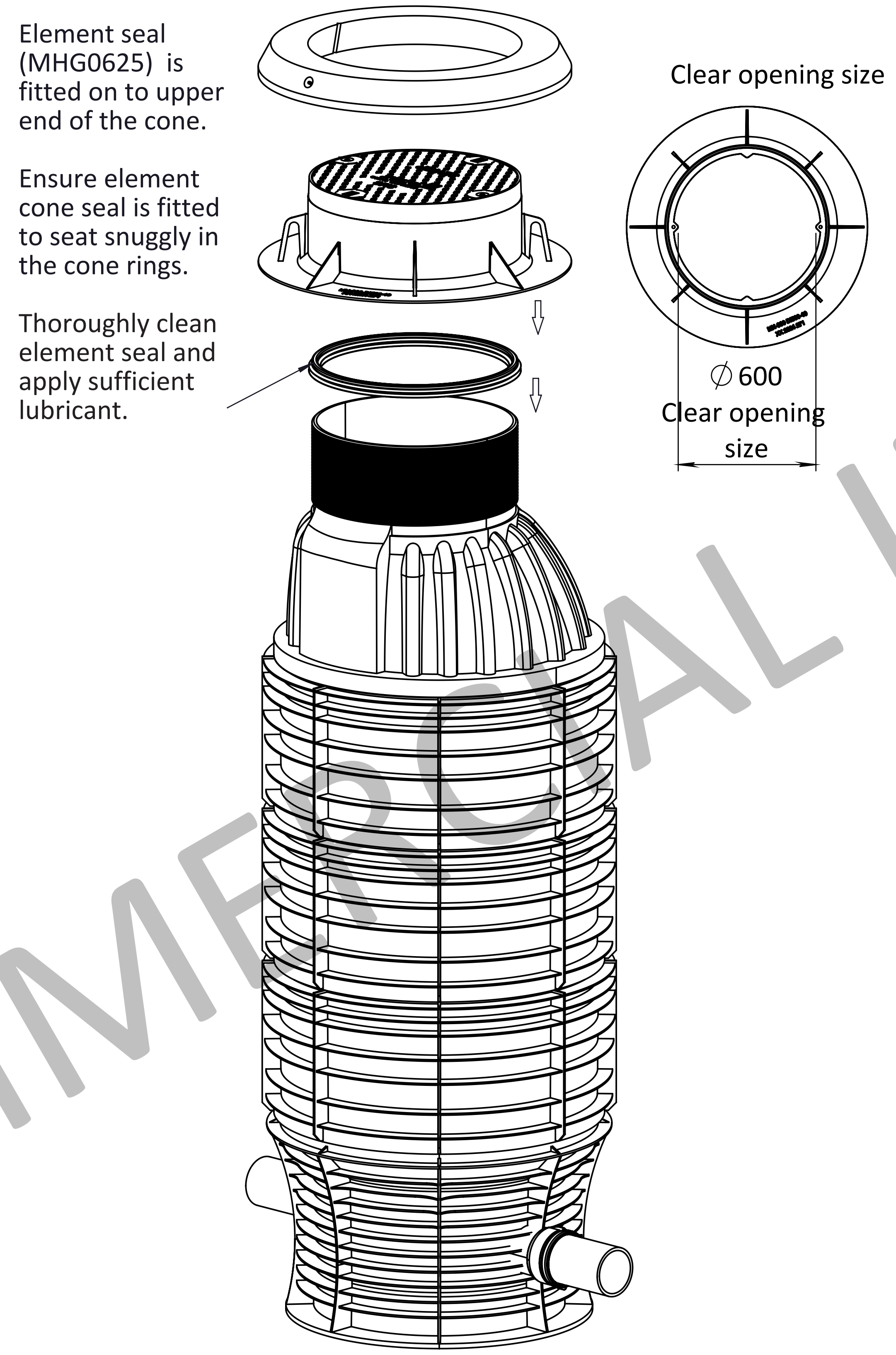
The SMS Class D Top Hat is designed to distribute traffic loads to the road foundation, preventing direct load transfer to the PP maintenance hole.

Before installation, apply the element seal MHG625 to the top of the PP maintenance hole and use sufficient lubricant. Assemble the Top Hat access cover horizontally and ensure it is centered over the maintenance hole on a stable, prepared base.

Important: There must be no direct load contact between the Top Hat and the PP maintenance hole.

The bedding for the Top Hat must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed.

For height adjustment of the access cover, refer to Step 6 (on Sheet 3).



OPTION 1: CONCRETE COVER PLATE INCORPORATING ACCESS COVER

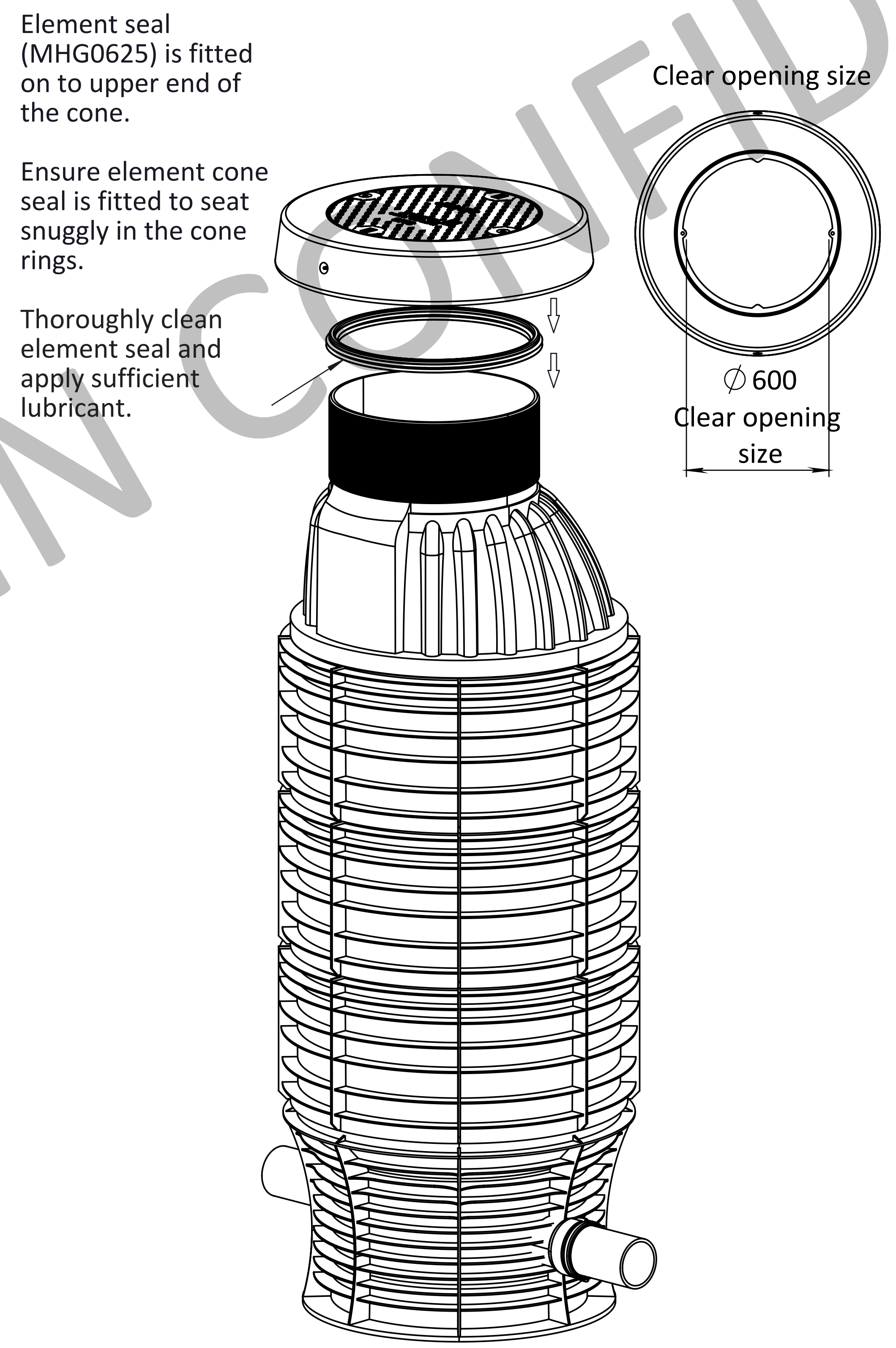
Install the maintenance hole in accordance with Steps 1 to 7 of the chamber assembly and installation procedure.

Apply element seal MHG625 to the top of the PP maintenance hole and use sufficient lubricant. Position the concrete cover plate horizontally and centrally over the maintenance hole on a stable, prepared base.

Important: Ensure there is no direct load contact between the concrete cover plate and the PP maintenance hole.

A commercial access cover (up to Class D) can be installed on top of the concrete cover plate. Height adjustment of the access cover can be achieved using concrete adjustment rings.

For further details on access cover height adjustment, refer to Step 6 (on Sheet 3).



OPTION 2: LOAD DISTRIBUTION RING MADE OF CONCRETE WITH ACCESS COVER

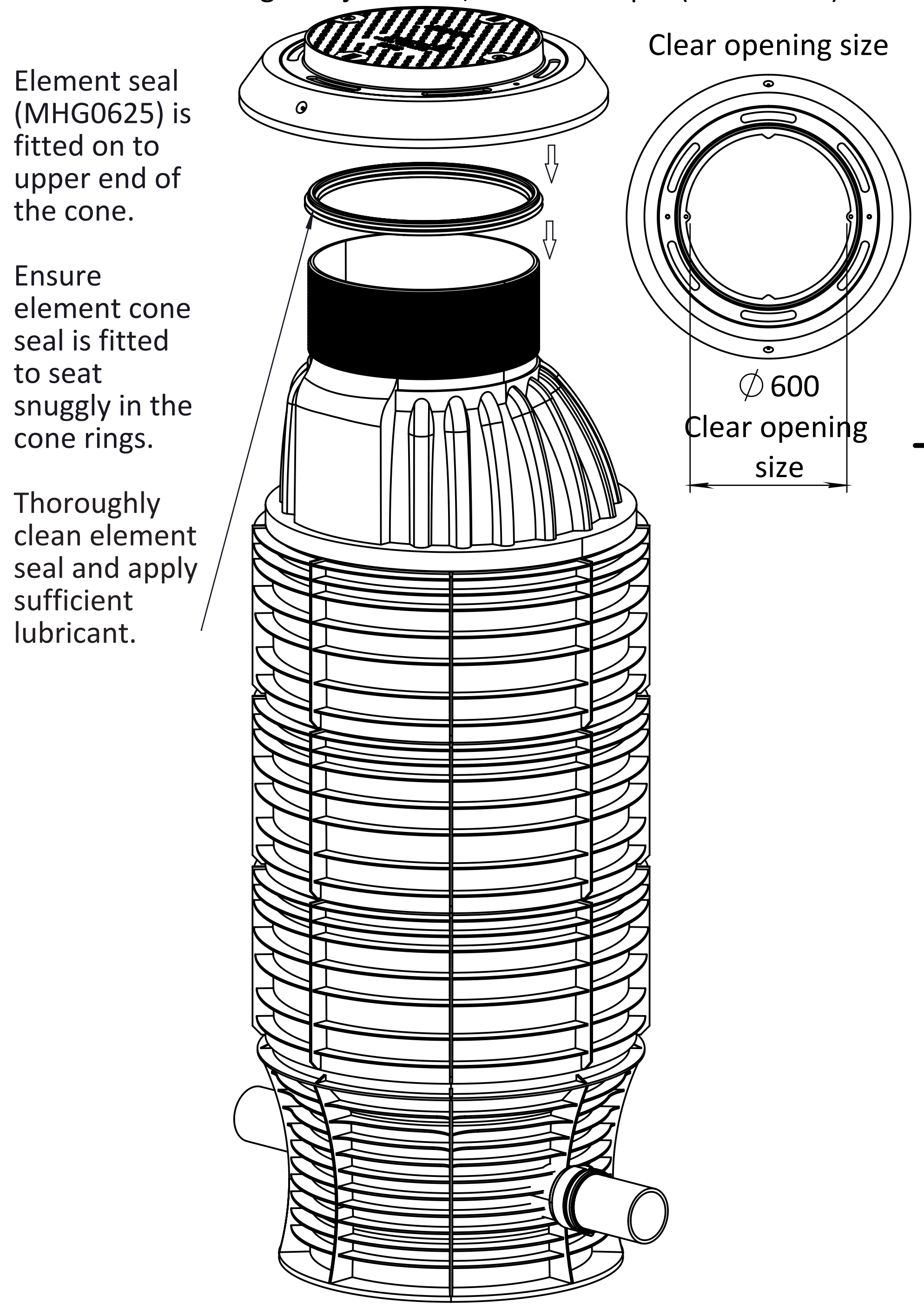
The SMS/ROMOLD concrete load distribution ring is designed to transfer traffic loads to the road foundation, preventing direct load on the PP maintenance hole.

Begin by assembling element seal MHG625 on top of the PP maintenance hole, applying sufficient lubricant. Position the load distribution ring horizontally and centered over the maintenance hole on a stable, prepared base.

Important: Ensure there is no direct load contact between the concrete ring and the PP maintenance hole.

Below the concrete load distribution ring (which extends approximately 40 mm above the edge of the upper unit), an EV2 modulus of at least 100 MN/m² must be achieved. The bedding must be level and free from point loads—granular material, sand, or concrete may be used as needed. If required, install the upper unit seal onto the upper unit neck before positioning the concrete ring, ensuring proper lubrication. The concrete ring must be centered during installation without disturbing the bedding. Until the access cover is installed, cover the concrete support ring with a steel plate. To achieve the required height, the cone neck can be trimmed along the corrugation lines.

For access cover height adjustment, refer to Step 6 (on Sheet 3)



On completion of installation and any required testing, access cover lid and bolting needs to be removed safely and DENSO Manhole sealing grease applied to the access cover seating area. Access cover lid to be re-installed and bolts tightened to 80-100Nm and caps plastic caps re-fitted.

SAFETY NOTE: Safety precautions and risk assessment need to be completed prior to this activity commencing as clear opening of access cover (600mm) will allow direct access into the maintenance hole.

IMPORTANT NOTE: Follow all installation guides including minimum gap allowances to ensure any loads are not directly transferred to the cone.

Material:		Description: **SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE**
Finish:		
Unless Otherwise Stated: All tolerances as stated; Surface Finish: 0.8μm; All Dimensions: mm. DECIMALS X ±1.0, X.X ±0.1, X.XX ±0.02; ANGLES X ±1°, X.X ±0.3°	Approx Weight:	Part Number:
Drawing Scale: NTS	DO NOT SCALE DRAWING IF IN DOUBT ASK	Drawing Number: MH-ASSY-DN1000-100; Sheet: 5 of 6; Revision: 5
Drawing Produced In Accordance With: AS 1100	sms	Drawn by: SB; Drawn Date: 11/03/25
Projection Method: THIRD ANGLE; Sheet Size: A2		Checked/Approved by: HB; Checked/Approved Date: 11/03/25

INSTALLATION DEPTH RANGE

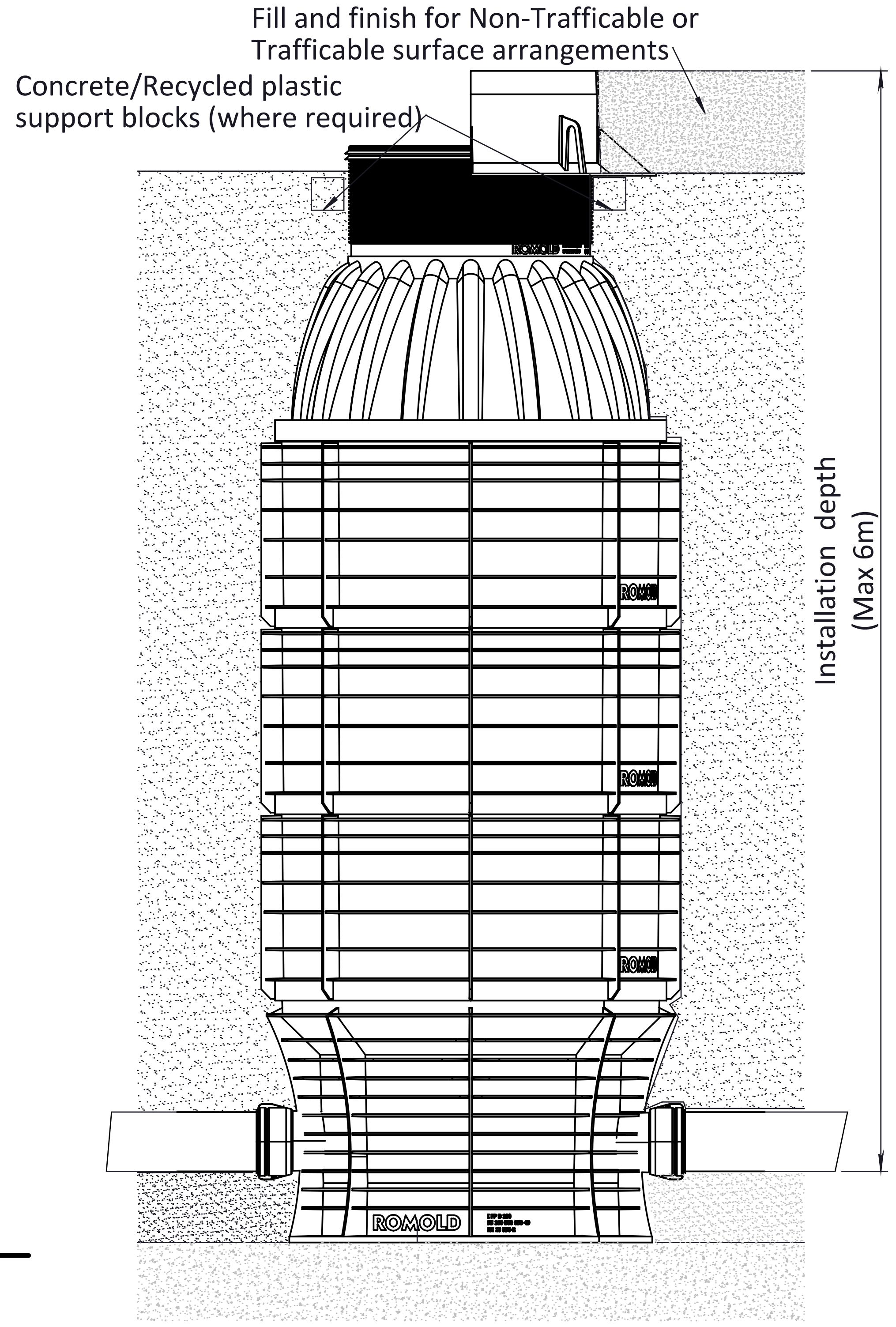


Table 1: Access Cover Top hat Installation Depths

Installation depth range ->	1500 1750	1750 2000	2000 2250	2250 2500	2500 2750	2750 3000	3000 3250	3250 3500	3500 3750	3750 4000	4000 4250	4250 4500	4500 4750	4750 5000	5000 5250	5250 5500	5500 5750	5750 6000
Component	Qty																	
Base	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Riser (250)	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0
Riser (500)	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
Riser (750)	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0
Riser (1000)	0	0	0	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4
Cone	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
DN600 MH Top hat Access Cover	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 2: Access Cover Option 1 & 2 Installation Depths

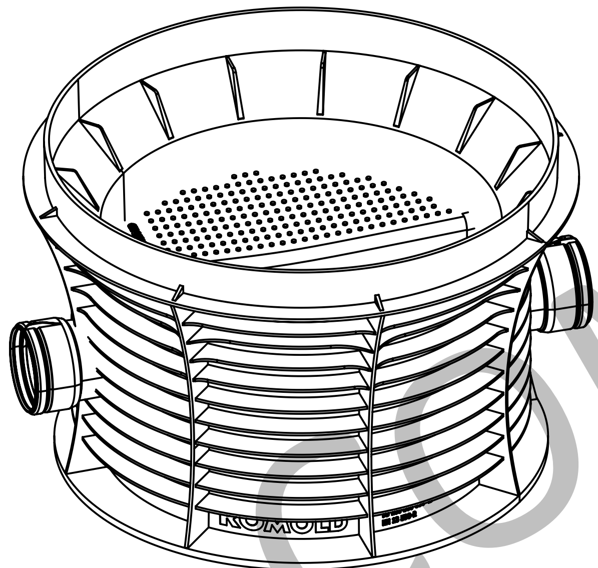
Installation depth range ->	1420 1670	1670 1920	1920 2170	2170 2420	2420 2670	2670 2920	2920 3170	3170 3420	3420 3670	3670 3920	3920 4170	4170 4420	4420 4670	4670 4920	4920 5170	5170 5420	5420 5670	5670 5920	5920 6170
Component	Qty																		
Base	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Riser (250)	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0
Riser (500)	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0
Riser (750)	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
Riser (1000)	0	0	0	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4	4
Cone	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
DN600 MH Access Cover option 1 or 2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Notes:
Other configurations of Riser components can be used to achieve same overall riser height.
Lubricant for Element Seals and DENSO Manhole sealing grease required for installation.

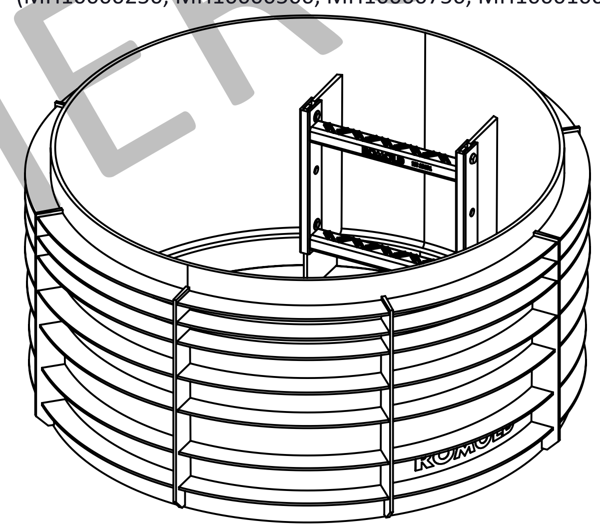
Weight Table

Component	Weight - KG
MH Base	50-65
MH Riser (250)	15
MH Riser (500)	25
MH Riser (750)	35
MH Riser (1000)	45
MH Cone	25
Standard: Top Hat Access Cover with Vegetation Ring	100 200
Option 1: Concrete Cover Plate incorporating Access Cover	200
Option 2: Concrete Load Distribution Ring with Access Cover	180

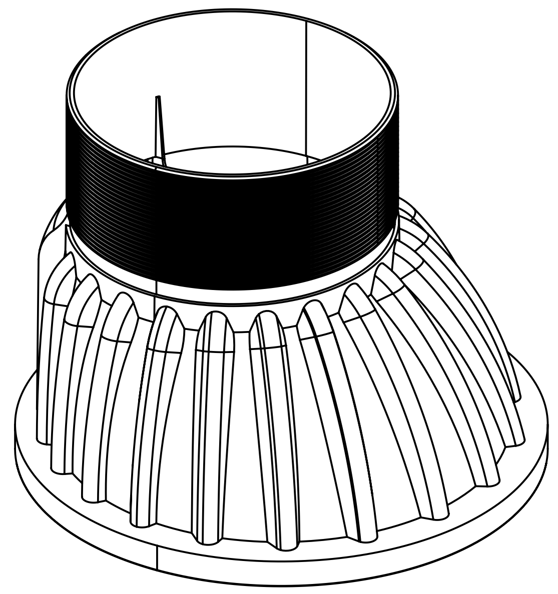
MH Base
refer table on Sheet 1 for configurations,
inlet/outlet sizes and codes



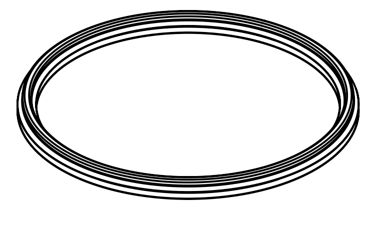
MH Riser
250mm/500mm/750mm/1000mm
(MH10000250, MH10000500, MH10000750, MH10001000)



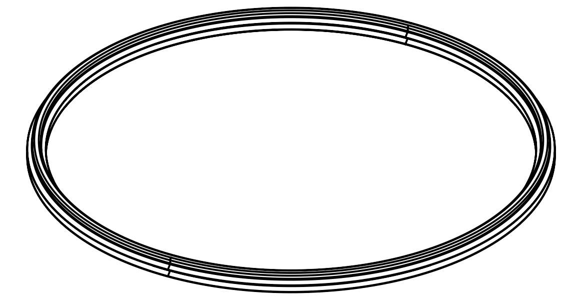
MH Cone
(MHRC600)



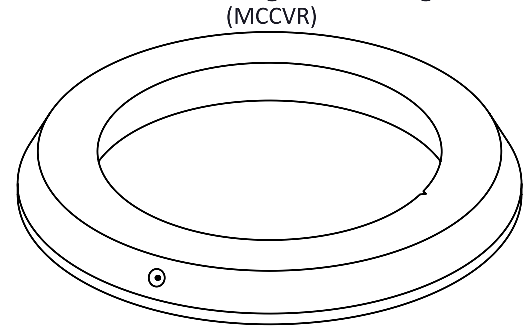
Element Seal
DN600 Cone/Access Cover
(MHG0625)



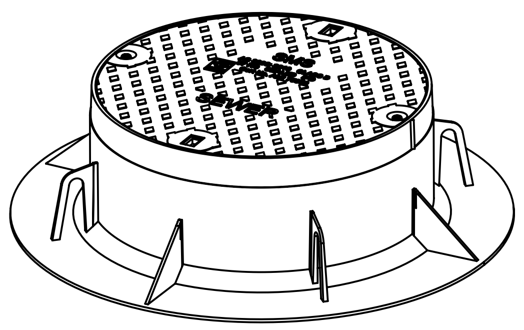
Element Seal
DN1000 Base/Riser/Cone
(MHG1000)



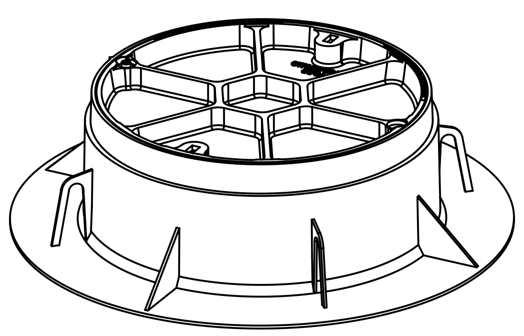
Concrete Vegetation Ring
(MCCVR)



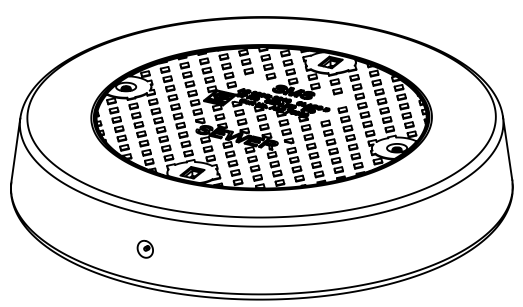
Standard: Top Hat Access Cover
(ACCMHTHCSDMC)



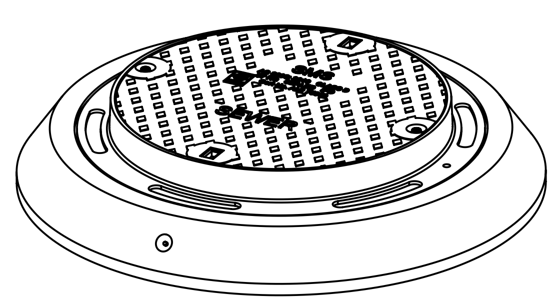
Top Hat Access Cover (Concrete Infill Lid)



Option 1: Concrete Cover Plate
incorporating Access Cover



Option 2: Concrete Load
Distribution Ring with Access Cover



ROMOLD

Unless Otherwise Stated: All tolerances as stated; Surface Finish: 0.8µm; All Dimensions: mm
DECIMALS: X ±1.0; X.X ±0.1; X.XX ±0.02
ANGLES: X ±1°; X.X ±0.3°

Drawing Scale: NTS
DO NOT SCALE DRAWING IF IN DOUBT ASK
Drawing Produced In Accordance With: AS 1100
Projection Method: THIRD ANGLE
Sheet Size: A2

Material:
Finish:
Approx Weight:

sms

Description: SMS/ROMOLD DN1000 POLYPROPYLENE (PP) MAINTENANCE HOLE
Part Number:
Drawing Number: MH-ASSY-DN1000-100
Sheet: 6 of 6
Revision: 5
Drawn by: SB
Drawn Date: 11/03/25
Checked/Approved by: HB
Checked/Approved Date: 11/03/25